

Marketing Campaign Analytics Portfolio Project

Portfolio Demonstration Project

This project was developed to demonstrate practical skills in data cleaning, exploratory data analysis, business intelligence, and interactive dashboard development using real-world marketing data.

Dataset

Bank Marketing Dataset

Technologies Used

Python (Pandas, Matplotlib, Seaborn)

Jupyter Notebook

Power BI

Project Objective

This portfolio project was created to showcase practical data analytics and business intelligence skills by analyzing a real-world bank marketing dataset.

The objective was to investigate customer responses to direct marketing campaigns, identify the factors influencing customer conversion, and transform raw data into meaningful business insights through exploratory data analysis and interactive dashboard development.

The project demonstrates the complete analytics workflow, including data preprocessing, feature engineering, visualization, KPI creation, and business recommendation generation using Python and Power BI.

Dataset Overview

A publicly available Bank Marketing dataset was used to simulate a real business analytics project.

The dataset contains approximately 11,162 customer records with 17 variables describing customer demographics, financial profiles, marketing campaign interactions, and the final campaign outcome. The attributes are grouped into three major categories:

- Customer Demographics
 - Age
 - Job
 - Marital Status
 - Education
- Financial Profile
 - Average Yearly Balance
 - Housing Loan
 - Personal Loan
 - Credit Default
- Campaign Details
 - Contact Method
 - Contact Day
 - Contact Month

- Call Duration
- Number of Contacts
- Previous Campaign Outcome

Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked and handled missing values.
- Removed duplicate records.
- Treated extreme outliers in balance and duration using the Interquartile Range (IQR) method.
- Encoded categorical variables.
- Created a numeric target column named **converted**.
 - Yes = 1
 - No = 0
- Grouped age into meaningful categories such as Young Adult, Middle-aged, and Senior.
- Generated Key Performance Indicators (KPIs):
 - Total Customers
 - Total Conversions
 - Overall Conversion Rate

These preprocessing techniques demonstrate common data preparation practices performed in real-world analytics projects before visualization and business reporting.

Exploratory Data Analysis

Several visualizations were created to understand customer behaviour and campaign effectiveness.

Customer Demographics and Occupation

Most customers belonged to the middle-age group (30–50 years). However, the highest conversion rates were observed among customers below 25 years and above 60 years.

Students and retirees recorded the highest subscription rates, whereas blue-collar workers and entrepreneurs demonstrated comparatively lower conversion rates.

Marital Status and Education

Single customers showed slightly higher conversion rates compared to married or divorced customers.

Customers with tertiary education consistently achieved better conversion performance, indicating stronger financial awareness.

Housing and Personal Loans

Customers without existing housing or personal loans demonstrated significantly higher conversion rates than customers with active loan obligations.

Communication Channel

Campaign effectiveness differed across communication methods. Cellular communication achieved considerably higher conversion rates than traditional telephone contact.

Monthly Campaign Performance

Although May recorded the highest number of customer contacts, it delivered one of the lowest conversion rates.

March, September, October, and December exhibited comparatively lower campaign volumes but substantially higher conversion performance.

Campaign Duration

Longer customer conversations, particularly those exceeding five minutes, showed a strong positive relationship with successful conversions.

Previous Campaign Outcome

Customers who had previously subscribed during earlier campaigns were much more likely to convert again, making historical campaign success one of the strongest predictive variables.

Key Performance Indicators

KPI	Business Purpose
Total Reach	Measures the total number of unique customers contacted.
Total Conversions	Represents customers who successfully subscribed to the term deposit.
Conversion Rate	Measures campaign efficiency by calculating the percentage of successful subscriptions.
Average Call Duration	Evaluates customer engagement and call effectiveness.

Interactive Dashboard Overview

To showcase business intelligence and reporting capabilities, an interactive Power BI dashboard was developed that allows users to explore campaign performance from multiple business perspectives.

The dashboard includes:

- Executive KPI Cards
- Demographic Segmentation
- Job-wise Conversion Analysis
- Education-wise Analysis
- Housing Loan vs Conversion
- Personal Loan vs Conversion
- Monthly Conversion Trends
- Call Duration Analysis
- Previous Campaign Outcome Analysis
- Interactive Slicers for:
 - Job
 - Month
 - Education
 - Marital Status
 - Loan Status

Strategic Recommendations

Based on the analytical findings, the following business recommendations were developed to demonstrate how data-driven insights can support marketing decision-making:

1. Prioritize students, retirees, and tertiary-educated customers.
2. Increase marketing activities during March, September, October, and December.
3. Re-target customers who responded positively to previous campaigns.
4. Focus on customers without existing housing or personal loans.
5. Improve call quality by encouraging longer consultative conversations.
6. Increase reliance on cellular communication and reduce dependence on landline campaigns.
7. Continuously monitor campaign performance through the interactive Power BI dashboard.

Conclusion

This portfolio project demonstrates an end-to-end data analytics workflow using Python and Power BI.

Starting from raw customer data, the project covers data cleaning, preprocessing, exploratory data analysis, feature engineering, KPI development, dashboard creation, and business insight generation. Rather than focusing solely on visualization, the project emphasizes how analytical techniques can be applied to solve real business problems and support strategic decision-making.

The interactive dashboard illustrates the ability to communicate complex analytical findings through intuitive visualizations, while the recommendations highlight how data can be transformed into actionable business strategies.

Overall, this project serves as a practical demonstration of skills in data analysis, business intelligence, storytelling with data, and dashboard development, making it suitable for inclusion in a professional analytics portfolio.